

NAVANEETHAKANNAN ARUMUGAM "Navaneeth"

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SUMMARY

Data Science professional skilled in real-time analytics, predictive modeling, domain-specific model adaptation, containerized deployment, and AI solutions delivering measurable business impact.

EXPERIENCE

AI Engineer Intern, Copani LLC, Florida, USA Jan'2025 - Present

Copani LLC delivers AI-powered agentic workflows for customer service automation

- Developed NLP pipelines using Hugging Face Transformers (primarily BERT; limited T5) to automate customer service understanding, boosting model accuracy and reducing human intervention.
- Designed API-driven workflows to automate email responses tailored to client-specific requirements, increasing processing efficiency while upholding strong data security and compliance measures.
- Engineered a finance-focused LLaMA3 model with LoRA fine-tuning in Ollama to analyze technical, fundamental, and sentiment data and deliver actionable stock insights.

Web Developer Intern, NSIC, Chennai, India July'2022 - Aug'2022

National Small Industries Corporation Limited

- Architected and launched 10+ responsive web pages using HTML and CSS, reducing bounce rates by 20%.
- Enhanced user experience through improved UI components, resulting in a 15% increase in user engagement.

PROJECTS

Small Business Index Analysis Insights and Forecasting [Data Engineering, PySpark, Streamlit]

- Developed an ETL workflow to ingest raw sales data, clean missing values, remove outliers, normalize ranges, and categorize features, transforming datasets into star-schema tables achieving 90% accuracy.
- Built supervised forecasting models for Fiserv Small Business Index across NAICS sectors to generate insights.
- Created a Streamlit analytics application delivering adjusted forecasts for 25+ subsectors, providing stakeholders with market indicators, interactive visualizations, and data-driven guidance for operational planning.

Airbnb Price Recommendation Model [MLOps, Docker, DagsHub, Streamlit]

- Processed data with SQLite3 and optimized ML models for 92% accurate Airbnb price prediction.
- Deployed containerized ML model using Docker; managed model storage and retrieval.
- Utilized DagsHub for version control and collaboration, and developed an interactive Streamlit app for real-time price exploration, enhancing decision making for hosts and travelers.

NYC Taxi Tipping Pattern Analysis [Data Analytics, Pandas]

- Analyzed 3 Million NYC taxi trips to uncover tipping patterns across passenger, trip, and socioeconomic factors.
- Identified tipping trends across payment types, fares, durations, additional charges, and borough income using exploratory analysis, generating insights that support operational planning and service optimization.

Face Recognition Check-In System [ML Engineering, OpenCV]

- Developed a real-time check-in system using Haarcascade detection and embedding-based identity classification.
- Optimized recognition decisions using embedding comparisons and tuned tolerance, maintaining robust performance under varied lighting, filtering unknown faces, and logging check-ins to Excel-based records.

INNOVATION & COMPETITION EXPERIENCE

Startup Boot Camp – Elysium Security, Buffalo, NY

Pitched an AI-driven surveillance concept using YOLO for real-time crime detection and automated threat alerts. Demonstrated strong innovation and market potential, winning 1st place at UB Startup Boot Camp.

EDUCATION

State University of New York at Buffalo, USA Jan'2024 - May'2025

M.S. in Engineering Science Data Science CGPA: 3.667/4

Annamalai University, Chidambaram, India July'2019 - May'2023

B.E. in Computer Science (AI & ML) CGPA: 8.95/10

SKILLS

Programming Languages: Python, JAVA, R, C/C++ , SQL, Matlab, Spark, Haskell
Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Statistics, Distributed Systems, Optimization Mathematics, Object-Oriented Programming, Data Mining Query Language, Embedded Systems
Framework: TensorFlow, Pytorch, Scikit-learn, Keras, Numpy, Pandas, Hadoop, Docker, Kubernetes
Platforms: Excel, MySQL, PostgreSQL, Power BI, Tableau, AWS, Git, Databricks, Neo4j